

Anomaly based detection

1. The research paper focuses on IDS which is working based on PCC – CNN i.e Pearson correlation coefficient for streamlined feature selection with Convolution Neural Network for traffic classification.
2. This approach has a superiority because the accuracy is 99.89 while maintaining a low false alarm rate.
3. This identifies both binary (Normal traffic vs malicious traffic) and multiclass attacks(DDOS, DOS, Probing, R2L and U2R)
4. Findings suggest that PCC – CNN is a lightweight and computationally efficient model.

What are Signature based and rule based?

Signature based are the ones which look for a specific match i.e the cryptographic hash or string or a key that matches a bad looking attack script or so.

Rule based are the ones which works based on the logic that is written by any of the experts and it keeps checking the condition and can validate various data and metadata at the same time and the main backdrop of this method is that it can go for a long amount of time if the logic is too big or the condition to be checked is quite big.

Linear feature extraction is a method where multiple features are compared together using PCC and if the PCC value is near to +1 then it stands that they are highly similar while 0 stands for no relationship and -1 stands for highly negative relationship.

EX : if two features has a PCC ~ 0.85 it means that they are likely similar and one of the feature is discarded.

PCC – CNN model is a state of the art model which is trained on 3 datasets NSL – KDD, CICIDS – 2017 and IOTID20 and compared with 5 other different ML models having an accuracy of 99.98% and a false alarm rate of 0.02%.

PCC is a filter based feature selection technique to identify highly valuable features which are then passe to CNN and is then checked for the binary and multiclass network anomaly classification.

Anomaly based creates a lot of false positives, therefore it is important to update it based on latest ML and AI models.

Commonly used datasets to design IDS are KDDCUP99, DARPA but the problem is that they have been created long ago and even the redundancy of the features between the classes

Weka is a tool that has different types of ML and DL classifiers to predict and compare how algorithms would perform.

1. NSL KDD : This dataset consists of 4 categories and the attacks can fall into any of them and they are DoS, R2L, U2R, Probe.
2. CICIDS2017 : This dataset consists of 4 categories and there are 11 types of simulated attacks that fall into these 4 categories and they are BENIGN, DoS, Firewall, Port Scan.
3. IOTID20 : uses raw packet files and this dataset has been purely created for academic purposes and this one has been categorized into 4 types Dos, Mirai, Scan, MITM ARP.

PIPELINE

It consists of

1. Data preprocessing : data is cleaned and all the unwanted features are removed and then the Null values /NaN values are replaced or deleted.
2. Feature Selection : It is very important as it increases the accuracy of the model and even prevents overfitting of the models and reduces the cost and training time of the model too, and this one does feature selection works on pearson correlation coefficient method and this method calculates the coefficient by evaluating the correlation between the instance and a class and to reduce overfitting we select the features contributing most to classification.
3. ML Algorithm : This consists of various ML algorithms that have been implemented and they are
 - a. Logistic Regression : This estimates the probability of a new event being a malicious one based on the previous events.
 - b. LDA : predicts the likelihood that a new set belongs to every class.
 - c. KNN : estimates how likely a point is to be a member of one or more groups
 - d. CART
 - e. SVM
4. PCC – CNN : This model builds an efficient IDS and this is a lightweight process because it will keep only the required data while discarding the whole unnecessary data. PCC is a filter based algorithm as well as feature selection technique. It even uses ReLU which is Rectified Linear Unit and this is an activation function and outputs the input directly if it is positive i.e if $x \geq 0$ then x is the output directly else it outputs 0.
5. Evaluation : This has been evaluated using Confusion matrix, Accuracy, Precision, F1-Score, False Alarm Rate, ROC (Receiver Operating Characteristics) – AUC(Area under Curve) Curve.